TECHNICAL REQUIREMENTS DOCUMENT (TRD)

Google Maps Business Analysis Platform

Version: 1.0

Project Type: Data Analytics Platform

Prepared By: Team ________

Submission: Week 3

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1. Technical Overview

The **Google Maps Business Analysis Platform** is a data analytics application designed to collect, clean, analyze, and visualize publicly available Google Maps business data. The platform processes business information to identify organizations with limited digital presence and presents insights through dashboards and reports.

The system follows a modular architecture where each component performs a specific task, ensuring scalability, maintainability, and efficient data processing.

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2. System Architecture

```text
Google Maps Business Data
        │
        ▼
```

```
Data Collection Module
      │
      ▼
Data Cleaning Module
      │
      ▼
Data Analysis Module
      │
      ▼
Visualization Dashboard
      │
      ▼
Report Generation Module
```

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3. Technology Stack

Component	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Dashboard	Power BI / Tableau
Dataset Format	CSV / Excel
IDE	VS Code / Jupyter Notebook

| Version Control | Git & GitHub |

| Documentation | Microsoft Word |

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4. System Modules

Module 1 – Data Collection

Purpose: Collect business information from Google Maps.

Input:

- Business Listings

Output:

- Raw Dataset

Collected Fields:

* Business Name
* Category
* Rating
* Reviews
* Website
* Address

* Phone Number

Module 2 – Data Cleaning

Purpose: Improve data quality before analysis.

Functions:

* Remove duplicate records
* Handle missing values
* Standardize categories
* Remove invalid entries

Output:

- Clean Dataset

Module 3 – Data Analysis

Purpose: Generate meaningful business insights.

Analysis Includes:

* Website Availability

* Rating Analysis
* Review Analysis
* Category Distribution
* Business Count
* Opportunity Identification

Module 4 – Dashboard

The dashboard displays:

* Total Businesses
* Businesses with Website
* Businesses without Website
* Average Rating
* Category-wise Analysis
* Charts & Graphs
* Business Opportunity Summary

Module 5 – Report Generation

Reports can be exported as:

* CSV
* Excel

* PDF

* PowerPoint Presentation

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5. Data Flow

```text
Business Data
      │
      ▼
Data Collection
      │
      ▼
Data Cleaning
      │
      ▼
Data Analysis
      │
      ▼
Dashboard
      │
      ▼
Report Export
```

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6. Data Validation Rules

* Business Name should not be empty.
* Duplicate records must be removed.
* Ratings should be between **1.0 – 5.0**.
* Review count cannot be negative.
* Missing website values should be marked as **”No Website”**.
* Missing fields should be handled appropriately before analysis.

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7. Performance Requirements

* Process datasets efficiently.
* Generate dashboards within a few seconds.
* Maintain high accuracy in calculations.
* Support datasets containing hundreds of business records.
* Ensure reliable report generation.

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8. Security Requirements

* Use only publicly available business information.
* Prevent accidental modification of the original dataset.
* Store processed data securely.
* Restrict unauthorized access to project files.

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9. Assumptions & Constraints

Assumptions

* Google Maps data is publicly available.
* Business details are reasonably accurate.
* Dataset is available in CSV or Excel format.

Constraints

* No live Google Maps API integration.
* Analysis depends on available dataset quality.
* Manual verification may be required for missing information.

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10. Testing Strategy

Test Type	Purpose
Data Validation Testing	Verify dataset accuracy
Functional Testing	Check all modules work correctly
Dashboard Testing	Verify charts and KPIs
Report Testing	Ensure successful export
Performance Testing	Measure processing speed

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11. Future Enhancements

* Google Maps API Integration
* AI-Based Business Opportunity Score
* Machine Learning Predictions
* Real-Time Dashboard
* Cloud Deployment
* Interactive Web Application
* Automated Lead Generation

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12. Conclusion

The **Google Maps Business Analysis Platform** is designed using a modular and scalable architecture that ensures efficient data collection, processing, analysis, and visualization. The selected technologies provide reliable performance while maintaining simplicity and flexibility. This technical design supports current project requirements and allows future enhancements such as AI integration, cloud deployment, and real-time analytics.
